



POWERBOX

ALL-IN-ONE POWER SOLUTION FOR YOUR CYBERDECK



COMPONENTS

- **USB-C PD SINK**
Requests 12V via PD protocol
- **TPS54620 ×2**
6A Buck DC-DC converters
- **IDEAL DIODES ×2**
Protects bucks from neg. voltage
- **DW01A BMS**
2S lithium protection IC
- **BQ258RGER**
Balance charger – up to 2A
- **PCB**
Custom KiCad design

PROJECT DESCRIPTION

PowerBox is a custom PCB power system designed for cyberdecks & portable embedded computers. It negotiates 12V over USB-C PD, runs dual TPS54620 buck converters for regulated output, and manages a 2S Li-Ion pack with balance charging & BMS

MADE BY:

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KiCad • Open Hardware

github.com/Alvin-Alford/PowerBox

[PCB files](#) • [BOM](#) • [Gerbers](#) • [Notes](#)



SCAN FOR REPO